

5) a) X is the # of correct problems

$n=20$ trials $p=0.85$

$X \sim (20, 0.85)$ Binomial Distribution

$$\begin{aligned} E[H_1] &= E\left[\frac{X}{20}\right] \quad H_1 = \frac{X}{20} \\ &= \frac{1}{20} E[X] \\ &= \frac{20 \cdot 0.85}{20} = 0.85 \end{aligned}$$

$$\text{Variance: } \text{Var}(H_1) = \text{Var}\left(\frac{X}{20}\right) = \frac{1}{20^2} \text{Var}(X)$$

$$= \frac{20 \cdot 0.85 \cdot 0.15}{400} = \frac{2.55}{400} = 0.006375$$

b) Average grade: H

Distribution:

$$H = \frac{1}{10} \sum_{i=1}^{10} H_i$$

We used iid, summation, because every question set is independent

$$E[H] = E[H_1] = 0.85$$

$$\text{Var}(H) = \text{Var}\left(\frac{1}{10} \sum_{i=1}^{10} H_i\right) = \frac{1}{100} \text{Var}\left(\sum_{i=1}^{10} H_i\right)$$

$$\text{Var}\left(\sum_{i=1}^{10} H_i\right) = \sum_{i=1}^{10} \text{Var}(H_i) = 10 \cdot 0.006375 = 0.06375$$

$$\text{Var}(H) = \frac{0.06375}{100} = \boxed{0.0006375}$$